BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Inputs, Processes and Outputs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is algorithmic thinking?

2. What does an encryption utility produce?
   A) A smaller file
   B) A sorted folder
   C) Ciphertext
   D) Extra storage space

3. What does a compression utility do?

4. A CPU reads an instruction at memory address 120. Correct both claims. Explain why. (a) MAR holds a calculation result. (b) MDR always holds the next instruction's address.

Today's objectives

Identify inputs, processing and output for a specified problem. Construct an accurate IPO diagram.

Key vocabulary - Read, cover, recall

Input
Data supplied to a program.

Processing
Operations or decisions performed on data.

Output
Result produced by an algorithm.

IPO
Input-Process-Output model.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Start with the result

Every problem asks for a useful result: a total, decision, message or control signal. Identify what the result should be first.

Identify specific inputs

Inputs are supplied values such as length and width, a sensor reading or a selected option. Name exact values, rather than saying 'some data'.

Describe the processing

Processing includes calculations, comparisons or decisions. For example: total = unitPrice x quantity, rather than 'the computer works it out'.

Output completes the task

Output could be a displayed cost or warning message. Check that it follows from the processing and answers the stated question.

I Do - Worked model

A program takes a rectangle's length and width and calculates its area. Identify the input, processing and output. [4 marks]

- The result required is the rectangle's area.
- Inputs are length and width [1], two values supplied to the algorithm [1].
- Processing multiplies length x width [1].
- Output displays the area [1]. A separate result follows the calculation.

We Do - Guided practice

1. A snack shop program calculates the total for five identical snacks. Identify input, processing and output. [4 marks]

2. A temperature monitor displays an ice warning when a reading is below 2°C. Explain input, processing and output. [4 marks]

3. A ticket machine accepts an adult fare and a child fare and displays their total. Give four ordered steps. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the line that prints the pixel count.

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
```

You Do - Independent exam questions

Q2. Define processing within an algorithm. [2 marks]

Q3. A streaming service calculates subscription cost from months purchased and a price per month. Describe the IPO stages. [4 marks]

Q4. Explain why identifying IPO accurately helps when designing a school uniform ordering algorithm. [6 marks]

Exit Ticket - True or False?

Output is produced after suitable processing has been performed on required inputs.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Developing a clear, ordered and unambiguous method for solving a problem.

2. C - Ciphertext

3. It reduces the storage space required for data, using a suitable lossless or lossy method.

4. (a) The MAR holds the address being accessed. (b) The MDR holds data or an instruction moving to or from memory. An address is different from the data at it.

I Do - Explain the model

The result required is the rectangle's area.

Inputs are length and width [1], two values supplied to the algorithm [1].

Processing multiplies length x width [1].

Output displays the area [1]. A separate result follows the calculation.

We Do - Model answers

1. Inputs: price per snack [1] and quantity 5 [1]. Process: multiply price x 5 [1]. Output: the total price [1].

2. Temperature from the sensor is input [1]. The program compares it with 2°C [1]. If below the threshold [1], it outputs an ice warning [1].

3. Input adult fare [1]. Input child fare [1]. Add them [1]. Display the sum [1].

Code First - Answer

```
imageWidth = 10
imageHeight = 8
pixels = imageWidth * imageHeight
print(pixels)
```

Expected output: 80

You Do - Self-mark

Q2: Operations/calculations performed on inputs [1] to obtain results [1].

Q3: Inputs are number of months [1] and monthly price [1]. Processing multiplies them [1]. Output displays the cost [1].

Q4: Inputs could be item types and quantities [1] and unit prices [1]. Processing multiplies unit prices and quantities [1] and totals the order [1]. Output gives total price/confirmation [1]. Precise IPO helps plan/test the algorithm [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.